

SAHIL RAJESH WARKE

Nashik, Maharashtra, India • rswarke1972@gmail.com • +91 9730415213 • [Portfolio \(HyperAgent & 60 Apps\)](#) • github.com/rswarke1972-art • [LinkedIn](#)

PROFESSIONAL SUMMARY

Computer Engineering student and systems researcher specializing in **speculative parallel scheduling algorithms for autonomous agentic AI** and prolific architect of **60 production web applications and simulation suites** deployed on GitHub Pages. Author of **HyperAgent** (patent-pending speculative P-DAG scheduling engine achieving 2.03x to 2.25x speedup with zero state contamination) and a published 276-page philosophical psychology novel on Internet Archive. Strong academic record (**8.24 CGPA** across 4 completed semesters) actively preparing for GATE CS/IT.

TECHNICAL SKILLS

Systems & Algorithms:	Speculative Parallel DAG Scheduling, Kahn's Wave Partitioning, Copy-on-Write State Trees, Deterministic Rollback, LRU Caching
Languages & Core:	JavaScript (ES6+), Python 3, C/C++, HTML5, CSS3, SQL
Frontend & Graphics:	React.js, Vite, HTML5 Canvas API, SVG Vector Paths, Progressive Web Apps (PWA), Web Audio API
Tools & Environment:	Git/GitHub, GitHub Pages, Google Cloud Platform (GCP), Linux/Windows, LocalStorage Engines

SYSTEMS RESEARCH & GROUPED ENGINEERING SUITES (60 DEPLOYED APPS)

- 1. Distributed Systems & AI Scheduling: HyperAgent** ([Patent-Pending Architecture](#), [IEEE Manuscript Draft](#)) [Live Algorithmic Workbench](#)
 - Formulated agent workflows as Probabilistic DAGs (P-DAG) using Kahn's topological concurrency wave partitioning to eliminate serial tool latency.
 - Implemented branch-isolated copy-on-write state trees with deterministic O(1) pointer rollback on prediction mismatch, mathematically proving $S_HyperAgent == S_Sequential$ across a formal 7/7 automated unit test suite.
 - Conducted parametric accuracy sweeps across prediction accuracy with cold cache, discovering an empirical break-even threshold ($\alpha^* = 40\%$) and achieving 2.03x to 2.25x speedup over sequential execution with sub-millisecond LRU caching (p50 = 0.0018ms).
- 2. Symmetrical Game Systems & Forensic Simulations** ([Project L.I.Z.](#), [Fractured Self](#), [LinguaWar](#), [Pawnova](#)) [Live on GitHub Pages](#)
 - Architected symmetrical multi-perspective game loops driven by dual synchronized state machines (Detective vs. Killer) sharing a canonical mutable crime scene model with Bezier tangent deduction corkboards.
 - Engineered an interrogation biometric engine with live ECG heart-rate simulation, composure decay, and zero-asset procedural audio generation via the browser Web Audio API.
- 3. World Linguistics & Writing Systems Suite** ([18 Platforms: GreekFlow, HebrewFlow, KhmerFlow, ArmenianFlow, etc.](#)) [18 PWA Applications](#)
 - Developed 18 specialized linguistics platforms decoding ancient and modern scripts (abugidas, alphabets, logographs) across Greek, Hebrew, Khmer, Amharic, Cyrillic, Georgian, and Burmese.
 - Integrated dynamic SVG stroke order vector animations, International Phonetic Alphabet (IPA) audio guides, Devanagari/Ge'ez ligatures, spaced-repetition drills, and offline PWA service workers.
- 4. Statecraft, Philosophy & Creative Studios** ([LeaderAtlas](#), [Conjectura](#), [Ashtavakra](#), [ManhwaMaker Studio](#)) [Zero-Dependency SPAs](#)
 - Built zero-dependency client-side single-page applications indexing 447+ world leaders across 199 nations and classical Sanskrit/Arabic philosophical traditions using tiered asynchronous JSON memoization.
 - Created a digital webtoon authoring workstation supporting multi-layer HTML5 Canvas rendering, dynamic panel geometry re-slicing, and vertical-scroll mobile export serialization.

PUBLISHED LITERARY WORK

"The Questions That Heal" ([Published Philosophical Psychology Work, 276 Pages](#)) [Internet Archive Public Access](#)
Authored a 276-page philosophical psychology work exploring memory, trauma recovery, emotional resilience, and self-inquiry through the journey of Professor Vinay across 10 structured therapeutic arcs. Preserved with permanent open-access indexing on Internet Archive.

EDUCATION & ACADEMIC PREPARATION

Bachelor of Engineering (B.E.) in Computer Engineering - MET's Institute of Engineering, SPPU, Pune **2024 - 2028 (Expected)**
Current Cumulative Grade Point Average (CGPA): **8.24 / 10.00** (Semesters I - IV completed) [Nashik, Maharashtra](#)
GATE CS/IT & Core Systems: Data Structures & Algorithms, Operating Systems, DBMS, Theory of Computation, Computer Networks, Discrete Mathematics.